

Fault-Tolerant Model Predictive Control of a Quadrotor with an Unknown Complete Rotor Failure

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Abstract: In recent years, with the wide application of quadrotors in critical areas, the safety issue has attracted more and more attention and fault-tolerant control with rotor failure has become a focus of research. A fault-tolerant control (FTC) method with fault detection and diagnosis (FDD) is proposed for a quadrotor with an unknown complete rotor failure. FTC uses nonlinear model predictive control (NMPC) to achieve stable control of the position in the case of a complete rotor failure. FDD uses concurrent learning to diagnose and parameterize the unknown failure. Different from traditional linear or cascaded nonlinear fault-tolerant control methods, the proposed method does not rely on the assumption of small perturbation linearization or time-scale separation. It can autonomously implement fault-tolerant control without any failure information in priori, and can consider and deal with nonlinear dynamics and actuator constraints caused by failures. Therefore, the method has a better control effect and system bandwidth. Through simulation, the proposed method is compared with the cascaded MPC method. Under the same conditions, the cascaded MPC crashes due to the saturation of the inner loop. In contrast, the proposed method can autonomously identify the failure when an unknown complete rotor failure occurs, and achieve stable control of the position by sacrificing yaw stability.

Key Words: Nonlinear model predictive control (NMPC), Concurrent learning, Fault-tolerant control (FTC), Quadrotor

1 Introduction

In recent years, unmanned aerial vehicles (UAVs), particularly quadrotors, have been very popular due to their mechanical simplicity, vertical take-off and landing (VTOL) capability, and high agility [1]. They are widely used in various complex areas such as package delivery, security inspection, and forest fire prevention [2–4]. However, the application in complex areas also highlights higher safety requirements for quadrotors. For example, in the process of urban package delivery, the quadrotor may collide with other aircraft or buildings and damage the propeller blades. And the simple structure determines that there is no actuator redundancy. This leads to decline in flight performance or even crashes when the actuator is damaged, which is dangerous for urban ground population [5]. Therefore, it is critical to research the fault-tolerant control (FTC) of a quadrotor with actuator failure.

The existing FTCs can be classified into passive and active approach [6]. The passive FTC uses robust control approaches to make the closed-loop control system insensitive to actuator failures. As a result, there is no need to adjust the parameters and structure of the controller when the failure occurs. In [7], chaotic particle swarm optimization (CPSO) is proposed to optimize the PID parameters of a quadrotor, which improves the flight performance and robustness, and achieve a partial passive FTC. In [8], the closed-loop stability condition of sliding mode control is investigated, and the results show that it is able to maintain the desired performance without failure information. On the contrary, the active FTC detects the failure condition through fault detection and diagnosis (FDD) and adjusts the parameters or structure of the controller adaptively. The literature [9] uses a gain-scheduled PID FTC method, which is designed with

two different parameters correspond to the normal case and the failure case, and achieve an active FTC in the case of partial actuator failure by switching parameters. In [10], a nonlinear robust adaptive fault-tolerant position controller is addressed, which can deal with the failure of multiple motors. Moreover, for a more critical and challenging situation with a complete actuator failure, which means that an actuator is unable to generate thrust, a pioneering work is proposed in [11]. It reveals that the maneuverability of position can be maintained by sacrificing the yaw axis stability when a complete rotor failure occurs. Based on this work, a variety of methods are presented. In [12], a linear, time-invariant description of deviations from the attitude equilibrium is derived, and it is used to design a linear quadratic regulator (LQR) FTC method for the complete loss of one, two, and three propellers. In [13], on the other hand, an incremental nonlinear dynamic inversion (INDI) method is used as an inner-loop nonlinear fault-tolerant controller, as well as can handle external disturbances and model uncertainties.

However, there are some limitations to the existing methods. Firstly, existing FTC design methods rely on multiple assumptions, which may not be satisfied under a complete rotor failure [14]. As a result, it leads to a decrease in control performance. On the one hand, the design of linear controller relies on the assumption of small perturbation linearization near the hover point and therefore cannot handle model nonlinearity when deviating from the hover point. However, when a complete rotor failure occurs, it needs to sacrifice the yaw axis stability to achieve stable control of position, which results in a forced rotation along the yaw axis. This causes serious nonlinear rotational dynamics, which leads to decreased performance of the linear fault-tolerant controller. On the other hand, nonlinear controller with cascaded structure needs to design inner and outer loops independently, which relies on the time-scale separation principle. The outer-loop controller cannot consider the actuator constraint when generating the inner-loop

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desired commands. The generated commands may not be executable for the inner-loop, which will lead to the saturation of the inner-loop controller and thus affect the control performance and system bandwidth. Further, when a complete rotor failure occurs, the actuator constraints are extremely tight and the bandwidth of the generated force is critically inadequate. In this condition, the inexecutable desired commands generated by the outer-loop may cause system instability and eventually lead to the crash of the quadrotor. Secondly, most of the existing FTC system focus on controller design and assume that failure information is well known, which is not consistent with the actual application scenario. In practice, because of the complex environment, the failure information is often not known in advance.

In order to solve the above problems, an FTC approach with FDD is proposed of a quadrotor with an unknown complete rotor failure. The proposed FTC is based on nonlinear model predictive control (NMPC). NMPC is a kind of receding horizon control (RHC) algorithm. This algorithm predicts the future state according to the current state and the full-state nonlinear model, and solves the optimal control sequence that minimizes the cost. The use of a full-state nonlinear model, on the one hand, determines that the method does not rely on the assumption of small perturbation linearization, thus the method can effectively handle the nonlinear rotational dynamics caused by failure. On the other hand, it determines that the method can control all states simultaneously without cascaded structure. Therefore, it does not rely on the time-scale separation principle, which allows the control system to effectively handle the strong coupling caused by failure and increases the control system bandwidth. Another advantage of NMPC is its capability to handle control and state constraints [15]. The failures and limitations encountered by the quadrotor, such as motor saturation, propeller damage, and attitude limitations, can be expressed as control and state constraints. Therefore, the proposed method can correctly consider the actuator constraints caused by failures when solving the control inputs, and avoid generating inexecutable control commands, which guarantees the control performance. The proposed FDD is based on concurrent learning. It can estimate the loss of effectiveness (LOE) of the quadrotor actuator using both historical and current data, which guarantees the convergence of parameters in the absence of continuous excitation. The LOE estimated by this method is used to detect, diagnose and parameterize the unknown failure in order to provide failure information to the FTC, which allows the FTC system to handle failures autonomously.

The remainder of this paper is organized as follows. Section 2 presents the modeling of the quadrotor and makes a special treatment for rotation along the yaw axis. Section 3 presents the FTC based on NMPC and the FDD based on concurrent learning. Section 4 shows the result of the simulation, which contains a comparison of the proposed method with the cascaded MPC. Finally, section 5 provides the conclusion.

2 Quadrotor Modeling

The structure of quadrotor is defined as shown in Fig. 1. In order to represent the motion of the quadrotor, the earth-fixed inertial frame $O_I x_I y_I z_I$ and the body frame $O_B x_B y_B z_B$

are defined. The position of the quadrotor in the earth-fixed inertial frame is expressed as $\mathbf{P} = [x, y, z]^T$. $\mathbf{V} = [v_x, v_y, v_z]^T$ denotes the velocity. $\Theta = [\phi, \theta, \psi]^T$ means the Euler angle. And $\Omega = [p, q, r]^T$ is the angular velocity in the body frame.

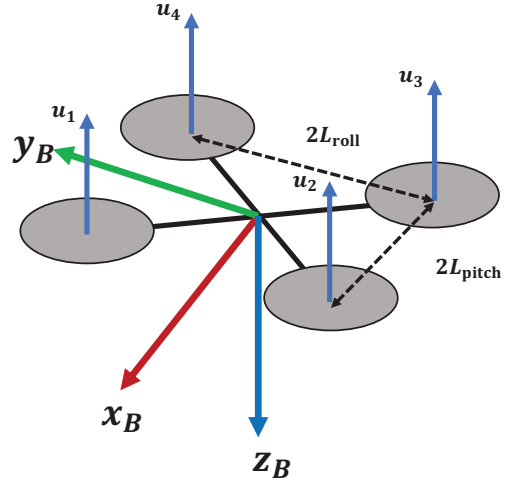


Fig. 1: Structure diagram of the quadrotor

The kinematic and dynamic models of the quadrotor are shown as follows.

$$\begin{aligned}\dot{\mathbf{P}} &= \mathbf{V} \\ m\dot{\mathbf{V}} &= m\mathbf{g} + \mathbf{R}_{IB}\mathbf{F} \\ \dot{\Theta} &= \mathbf{C}_{IB}\Omega \\ \mathbf{J}\dot{\Omega} &= -\Omega \times (\mathbf{J}\Omega) + \mathbf{M}\end{aligned}\quad (1)$$

Where, m and $\mathbf{J} = \text{diag}(J_{xx}, J_{yy}, J_{zz})$ denote the mass and inertia matrix of the quadrotor respectively. $\mathbf{R}_{IB}$ is the rotational matrix from the body frame to the inertial frame. $\mathbf{C}_{IB}$ is the matrix that transforms the angular velocity in the body frame to the derivative of the Euler angle. $\mathbf{F}$ and $\mathbf{M}$ are the force and the moment in the body frame, shown as

$$\mathbf{F} = \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix} + \mathbf{F}_a \quad (2)$$

$$\mathbf{M} = \mathbf{M}_c + \mathbf{M}_r + \mathbf{M}_a \quad (3)$$

where $\mathbf{M}_r$ denotes the moment generated by the gyro effect [16]. $\mathbf{F}_a$ and $\mathbf{M}_a$ are aerodynamic force and moment respectively, which is difficult to model accurately. T and $\mathbf{M}_c$ are control force and moment respectively, which is generated by rotor.

$$\begin{pmatrix} T \\ \mathbf{M}_c \end{pmatrix} = \begin{bmatrix} \mathbf{B}_{T\omega} \\ \mathbf{B}_{M\omega} \end{bmatrix} \lambda \mathbf{u} = \mathbf{B}_{\nu\omega} \lambda \mathbf{u} \quad (4)$$

Where, $\mathbf{u} = [\omega_1^2, \omega_2^2, \omega_3^2, \omega_4^2]^T$ means the rotor speed. $\lambda = \text{diag}(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ denotes the LOE of the actuator, which is used to parameterize the failure condition. $\lambda_i = 1$ means there is no failure and $\lambda_i = 0$ means the rotor is completely

damaged.

$$\mathbf{B}_{\nu\omega} = \begin{bmatrix} -k_T & -k_T & -k_T & -k_T \\ -k_T L_{\text{roll}} & k_T L_{\text{roll}} & k_T L_{\text{roll}} & -k_T L_{\text{roll}} \\ k_T L_{\text{pitch}} & k_T L_{\text{pitch}} & -k_T L_{\text{pitch}} & -k_T L_{\text{pitch}} \\ k_M & -k_M & k_M & -k_M \end{bmatrix} \quad (5)$$

where L_{roll} and L_{pitch} are the arm of roll and pitch respectively. k_T and k_M are the coefficient of thrust and torque.

In [11], it is pointed out that the equitability of the full-state can not be maintained with a complete rotor failure. But the controllability of position can still be maintained by sacrificing the stability of the yaw axis, which is called the reduced control or yaw spinning strategy. However, it raises a higher requirement for the modeling of the quadrotor. In the modeling process, a 3-2-1 sequential Euler angular rotation is generally used to describe attitude motion. As a result, there is

$$\mathbf{R}_{IB} = \mathbf{R}_z^{-1}(\psi) \mathbf{R}_y^{-1}(\theta) \mathbf{R}_x^{-1}(\phi) \quad (6)$$

where

$$\begin{aligned} \mathbf{R}_x(\phi) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix} \\ \mathbf{R}_y(\theta) &= \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} \\ \mathbf{R}_z(\psi) &= \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned} \quad (7)$$

Finally, the component of thrust in each axis is written as

$$\begin{aligned} \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix} &= \mathbf{R}_{IB} \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix} \\ &= \begin{bmatrix} T(\cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi) \\ T(\cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi) \\ T \cos \phi \cos \theta \end{bmatrix} \end{aligned} \quad (8)$$

From Eq. (8), it shows that the thrust vector expressed in 3-2-1 sequential Euler angular rotation will be affected by yaw motion. Consequently, ϕ and θ need to change rapidly with ψ to maintain a constant thrust vector expression under the yaw spinning strategy. To solve this problem, a modeling method using 1-2-3 sequential Euler angular rotation is proposed in [17] such that

$$\mathbf{R}_{IB} = \mathbf{R}_x^{-1}(\phi) \mathbf{R}_y^{-1}(\theta) \mathbf{R}_z^{-1}(\psi) \quad (9)$$

The component of thrust in each axis is written as

$$\begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix} = \begin{bmatrix} T \sin \theta \\ -T \cos \theta \sin \phi \\ T \cos \phi \cos \theta \end{bmatrix} \quad (10)$$

From Eq. (10), it shows that yaw no longer affects the thrust vector expression.

3 Fault-Tolerant NMPC with FDD

An FTC approach is designed for the case of an unknown complete rotor failure, whose structure diagram is shown as Fig. 2. Fault-tolerant NMPC is used to control the quadrotor according to the failure condition. Concurrent learning is used to detect and diagnose the unknown failure condition and provides the failure information to NMPC.

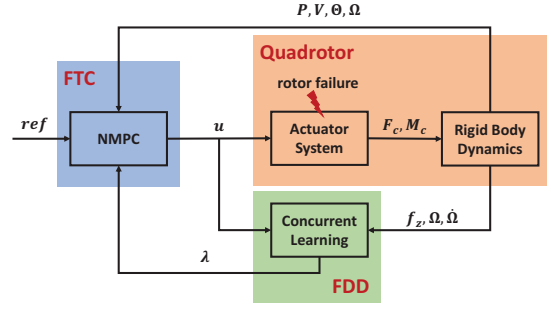


Fig. 2: Structure diagram of FTC system

3.1 Fault-Tolerant NMPC

Model predictive control (MPC) is a kind of receding horizon control (RHC) algorithm. This algorithm takes the current state as the initial state and predicts the state sequence of N steps in the future according to the model and control input sequence. Then it solves a control input sequence that minimizes the cost and takes the first item as the current control input. This process is executed again at the next step.

The quadrotor dynamics Eq. (1) can be expressed in the form of a general nonlinear system.

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t)) \quad (11)$$

It can be further written as a discrete form.

$$\mathbf{x}(t+1) = \mathbf{F}(\mathbf{x}(t), \mathbf{u}(t)) \quad (12)$$

Where, $\mathbf{x}(t)$ denotes the system state vector $\mathbf{x} = [\mathbf{P}, \mathbf{V}, \mathbf{\Theta}, \mathbf{\Omega}]^T$ at time t . $\mathbf{u}(t)$ denotes the system control input vector $\mathbf{u} = [\omega_1^2, \omega_2^2, \omega_3^2, \omega_4^2]^T$ at time t .

NMPC takes $\mathbf{u}_{k:k+N-1}$ as the optimization variable and minimizes the following cost function

$$\begin{aligned} \min_{\mathbf{u}_{k:k+N-1}} & \|\mathbf{x}_N - \mathbf{x}_{ref}\|_{\mathbf{P}}^2 \\ & + \sum_{i=k}^{k+N-1} \|\mathbf{x}_i - \mathbf{x}_{ref}\|_{\mathbf{Q}}^2 + \|\mathbf{u}_i - \mathbf{u}_{ref}\|_{\mathbf{R}}^2 \end{aligned} \quad (13)$$

where k is the current time step and N is the prediction horizon. $\|\mathbf{x}\|_{\mathbf{P}}^2 \triangleq \mathbf{x}^T \mathbf{P} \mathbf{x}$ and the positive-definite matrix $\mathbf{Q}$ and $\mathbf{R}$ are the process cost weight matrix of state and control input respectively. $\mathbf{P}$ is the terminal cost weight matrix to guarantee the stability of the system, which is usually obtained by solving the discrete-time Riccati equation [18].

In order to make the state change according to the dynamics, dynamic constraints Eq. (12) are added to the optimization problem. In order to satisfy the state constraints, such as the no-fly zone and flight envelope, state constraints are added to the optimization problem.

$$\mathbf{x}_i \in \mathbb{X} \quad (14)$$

In order to satisfy the physical constraints caused by the actuator and failure, control input constraints are added to the optimization problem.

$$\mathbf{u}_{\min} \leq \mathbf{u}_i \leq \lambda \mathbf{u}_{\max} \quad (15)$$

Where, $\mathbf{u}_{\max}$ and $\mathbf{u}_{\min}$ are the upper and lower bounds of the physical constraints of the actuator, respectively. λ is the LOE of the actuator, which is explained in detail in Eq. (4). Finally, a terminal constraint is added to the optimization problem to guarantee the stability [18].

$$\mathbf{x}_N \in \mathbb{X}_f \quad (16)$$

Combining Eq. (12)-(16), a constrained optimization problem is obtained.

$$\begin{aligned} \min_{\mathbf{u}_{k:k+N-1}} & \|\mathbf{x}_N - \mathbf{x}_{ref}\|_P^2 \\ & + \sum_{i=k}^{k+N-1} \|\mathbf{x}_i - \mathbf{x}_{ref}\|_Q^2 + \|\mathbf{u}_i - \mathbf{u}_{ref}\|_R^2 \\ \text{s.t. } & \mathbf{x}_{i+1} = F(\mathbf{x}_i, \mathbf{u}_i) \\ & \mathbf{x}_i \in \mathbb{X} \\ & \mathbf{x}_N \in \mathbb{X}_f \\ & \mathbf{u}_{\min} \leq \mathbf{u}_i \leq \lambda \mathbf{u}_{\max} \end{aligned} \quad (17)$$

To solve this problem, it should be constructed through a multiple shooting method and solved by a sequential quadratic program (SQP) [19].

When a rotor failure occurs, λ in Eq. (15) needs to be modified according to the failure condition estimated by FDD. It makes NMPC aware of the failure condition of each rotor. As a result, it can fully consider the constraints caused by failures in the control process to avoid generating control commands that cannot be executed by failure rotor. Moreover, when a complete rotor failure occurs, in order to achieve stable control of position, it have to actively sacrifice the yaw axis stability, which can be achieved by setting the relevant terms Q_ψ and Q_r in $\mathbf{Q}$ in Eq. (13) to 0. In summary, NMPC can achieve active FTC under different failure conditions by simply adjusting λ and $\mathbf{Q}$.

3.2 FDD based on Concurrent Learning

In order to achieve FTC, NMPC needs real-time loss of effectiveness (LOE) information. An approach based on concurrent learning is proposed to estimate the LOE in [20]. This approach can make full use of historical data, which guarantees the convergence of parameters without continuous excitation. Inserting Eq. (4) into Eq. (1), the identification model is written as

$$\dot{\hat{\Omega}} = -J^{-1}(\Omega \times J\Omega) + J^{-1}B_{M\omega}u\hat{\lambda} - L(\hat{\Omega} - \Omega) \quad (18)$$

where $(\hat{\bullet})$ denotes the estimation value and L denotes the Luenberger gain.

Subtract Eq. (18) from Eq. (1)

$$\dot{\tilde{\Omega}} = -L\tilde{\Omega} + J^{-1}B_{M\omega}u\tilde{\lambda} \quad (19)$$

where $(\tilde{\bullet}) \triangleq (\hat{\bullet}) - (\bullet)$

In order to find an update law for $\hat{\lambda}$ that can guarantee Lyapunov stability, let

$$V = \frac{1}{2}\tilde{\Omega}^T \tilde{\Omega} + \frac{1}{2}\tilde{\lambda}^T \Gamma^{-1} \tilde{\lambda} \quad (20)$$

where Γ is a positive-definite matrix. The derivative of Eq. (20) is written as

$$\dot{V} = -\tilde{\Omega}^T L \tilde{\Omega} + \tilde{\Omega}^T J^{-1} B_{M\omega} u \lambda + \tilde{\lambda}^T \Gamma^{-1} \dot{\tilde{\lambda}} \quad (21)$$

Assuming that λ will not change in a short time, let $\dot{\tilde{\lambda}} = \dot{\lambda}$.

In order to guarantee Lyapunov stability, Eq. (21) need to be negative-definite. Let

$$\mathbf{y} = \begin{pmatrix} \dot{\Omega} + J^{-1}(\Omega \times J\Omega) \\ f_z \end{pmatrix} = \underbrace{\begin{pmatrix} J^{-1}B_{M\omega} \\ -\frac{1}{m}B_{T\omega} \end{pmatrix}}_{:=B_{y\omega}} \mathbf{u} \lambda \quad (22)$$

Where, f_z and Ω are the specific force and angular rate in the body frame respectively and can be measured by the sensor. A final update law of $\hat{\lambda}$ is shown as

$$\dot{\hat{\lambda}} = \dot{\lambda}_{CL} - \Gamma \mathbf{u}^T B_{M\omega}^T J^{-T} \tilde{\Omega} \quad (23)$$

The update law can be divided into two parts, the latter part is the excitation from current data, and the former part is the excitation from historical data.

$$\begin{aligned} \dot{\lambda}_{CL} &= -\Gamma \sum_{i=1}^p \mathbf{u}_{f,i}^T B_{y\omega}^T (B_{y\omega} \mathbf{u}_{f,i} \hat{\lambda} - \mathbf{y}_{f,i}) \\ &= -\Gamma \underbrace{\sum_{i=1}^p \mathbf{u}_{f,i}^T B_{y\omega}^T B_{y\omega} \mathbf{u}_{f,i}}_{:=H} \tilde{\lambda} \end{aligned} \quad (24)$$

A data selection and recording method is proposed in [21], which can guarantee that the matrix H in Eq. (24) is positive-definite. And then a following negative-definite expression is obtained by inserting Eq. (23) into Eq. (21)

$$\dot{V} = -\tilde{\Omega}^T L \tilde{\Omega} - \tilde{\lambda}^T H \tilde{\lambda} \quad (25)$$

Finally, it is proved that the update law Eq. (23) can make the estimation value $\hat{\lambda}$ converge to the truth value λ exponentially.

4 Simulation

In order to verify the validity of the proposed FTC approach, a numerical simulation is carried out. The physical parameters of the quadrotor used in the simulation is shown in Table 1. A reference trajectory is written as $\mathbf{P}_{ref} = \{\cos(t), \sin(t), 0.5t\}$. It is assumed that the Rotor 2 completely fails after $t = 5s$ and the FTC system has no prior information about the failure. The implementation of the FTC is based on ACADOS [22] and CasADi [23], whose parameters are shown in Table 2.

Table 1: Parameters of the Quadrotor

Parameters	Value
m	1.121kg
J_{xx}	0.0100kg · m ²
J_{yy}	0.0082kg · m ²
J_{zz}	0.0148kg · m ²
L_{roll}	0.1068m
L_{pitch}	0.0879m
k_T	2.0784×10^{-8} N/rpm ²
k_M	2.5648×10^{-10} N · m/rpm ²

The results of tracking reference trajectory and tracking error are shown in Fig. 3-4. In order to show the advantages of the full-state NMPC proposed in this paper, a cascaded

Table 2: Parameters of the FTC

Parameters	Value
t_{step}	0.004s
N	30
Q	$\text{diag}(10^3 \cdot \mathbb{I}^{3 \times 3}, 10 \cdot \mathbb{I}^{3 \times 3}, 10^2 \cdot \mathbb{I}^{3 \times 3}, \mathbb{I}^{3 \times 3})$
R	$10 \cdot \mathbb{I}^{4 \times 4}$
Γ	$10^{-3} \cdot \mathbb{I}^{4 \times 4}$

MPC method using the same parameters is added to the simulation for comparison. From Fig. 3-4, it can be seen that both NMPC and cascaded MPC can follow the reference trajectory in the case of no failures. However, since the outer loop of the cascaded MPC cannot consider the actuator constraints, the generated control commands cannot be fully executed by the inner loop. Therefore, the cascaded MPC generates a larger tracking error compared to the NMPC. When a rotor failure occurs, the actuator constraint is tighter, the system maneuverability is significantly reduced and the inner loop is more prone to saturation. Therefore, this defect is even more fatal in this case. From Fig. 3, it can be seen that when a complete rotor failure occurs, the outer loop of cascaded MPC still generates the desired command as normal because it cannot consider the effects of the failure. The inner-loop MPC could not solve for a feasible solution under this desired command, which eventually leads to the crash of the quadrotor. In contrast, NMPC using a full-state nonlinear model does not rely on the time-scale separation assumption and can fully consider actuator constraints caused by failure. Therefore, it can continue to follow the reference trajectory. Fig. 4(b) and Fig. 8 show that, in the time since the failure occurred ($t = 5s$), due to the fact that the FDD has not yet converged, NMPC does not have access to sound failure information, so the tracking error has a large increase compared to the normal state. At around $t = 6s$, LOE estimated by FDD converges completely and the tracking error gradually recovers to a level similar to the normal state. In summary, compared with cascaded MPC, the proposed FTC method based on full-state NMPC can follow the reference trajectory better under normal conditions. After a complete rotor failure occurs, the proposed FTC method can effectively compensate for the effect of failure and maintains the ability to follow reference trajectory.

Fig. 5-6 show that although it is impossible to maintain the equitability of the full-state for the quadrotor with a complete rotor failure, the body z -axis direction can be controlled by sacrificing the stability of the yaw axis. Since the thrust direction of the quadrotor is only related to the body z -axis direction and irrelevant to the yaw, it guarantees that the quadrotor can generate the desired three-axis acceleration, which is the essence and foundation of the FTC with a complete rotor failure. Besides, Fig. 5 shows that in the early stage of the failure, the following of z -axis direction is inferior, which coincides with the results of large tracking errors in Fig. 4(b). While when FDD gradually converges, the body z -axis direction wobbles periodically along the reference z -axis direction. This control strategy is more energy-efficient [12], which is very effective for the quadrotor with a complete rotor failure because of the thrust shortage.

Fig. 7 is the result of the normalized control output of NMPC. It can be seen that, at the moment of complete fail-

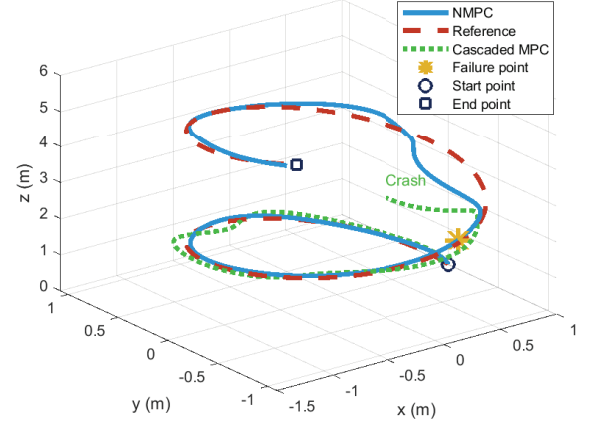
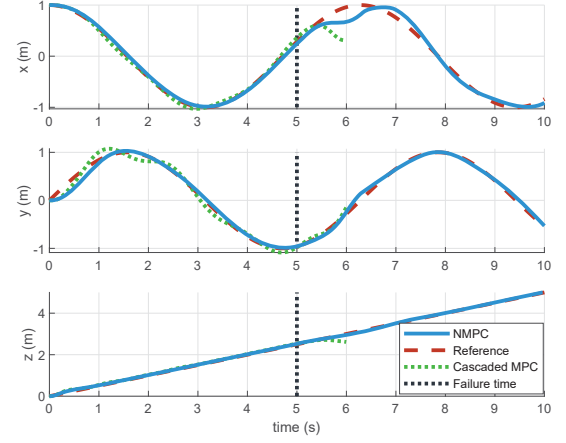
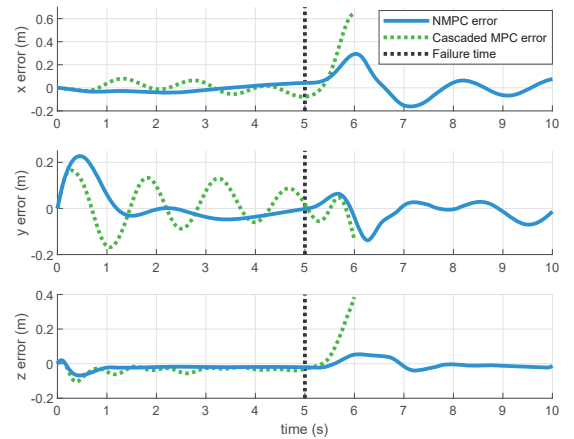


Fig. 3: The 3D tracking trajectory with a complete rotor failure. The blue solid line is the tracking trajectory of NMPC. The green dashed line is the tracking trajectory of a cascaded MPC using the same parameters, which crashes after failure occurs.



(a) Position tracking



(b) Tracking error

Fig. 4: Time response of position with a complete rotor failure

ure of Rotor 2 ($t = 5s$), since the FTC is not aware of the

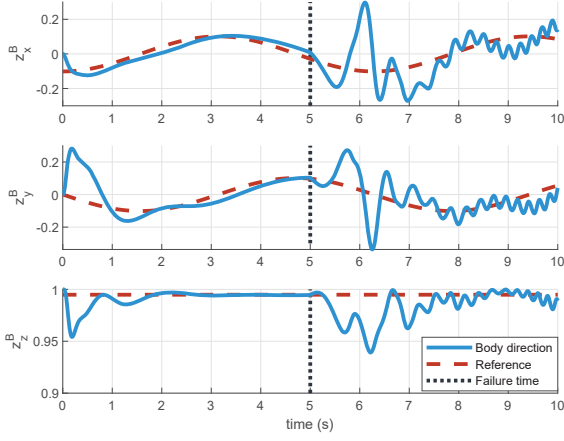


Fig. 5: Time response of body z -axis direction

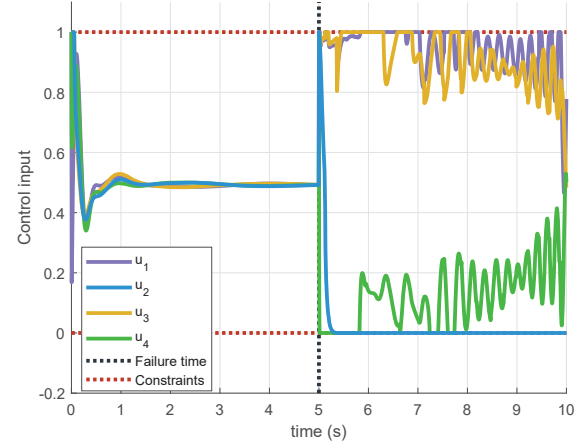


Fig. 7: Time response of control input

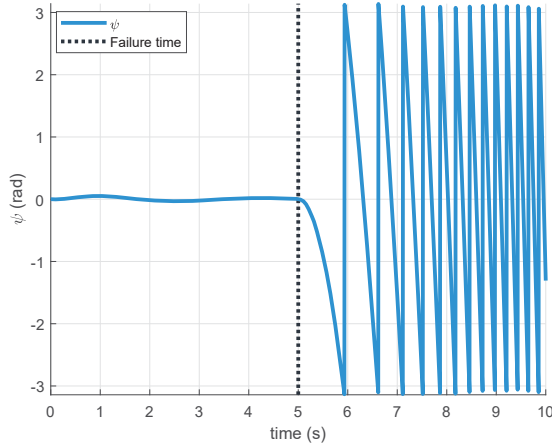


Fig. 6: Time response of yaw angle ψ

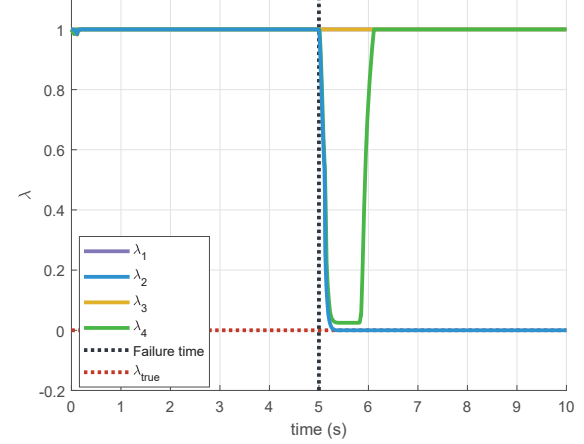


Fig. 8: Estimation of LOE with a complete rotor failure

failure, it only realizes the lack of thrust in Rotor 2 based on the state feedback and tries to compensate by increasing u_2 . At the same time, the u_4 of Rotor 4, which is opposite to Rotor 2, is gradually closed to achieve balance, and u_1 and u_3 are raised to compensate for the lack of thrust. However, with the gradual convergence of LOE estimated by FDD, the FTC realizes the failure of Rotor 2. Therefore, it gradually shuts down control of Rotor 2 and resumes control of Rotor 4 under the yaw spinning strategy. Throughout the regulation and optimization process, the control output fully considers the physical constraints of the actuators. This hard constraint ensures that the control commands generated by the control system can be fully executed by the actuators, which in turn guarantees the safety of FTC system.

Fig. 8 shows the time response of the LOE estimated by FDD. It can be seen that, after failure occurs, the λ_2 converges rapidly to 0, which provides fast and accurate failure information for FTC. And since the control system shut down u_4 at the beginning of the failure, the λ_4 also tends to decrease. However, as the control of Rotor 4 is recovered, the λ_4 converges to normal state. In summary, after an unknown complete rotor failure occurs, concurrent learning can automatically and rapidly estimate the LOE and communicates this information to NMPC, which achieves FTC

without any information in priori.

5 Conclusion

This paper proposes an FTC approach with FDD to achieve the fault-tolerant control of a quadrotor with an unknown complete rotor failure. In this approach, FTC based on full-state NMPC guarantees that it does not rely on assumptions of small perturbation linearization and time-scale separation. Therefore, it can consider and handle the nonlinear rotational dynamics, actuator constraints and strong coupling caused by failures, which increases the control system bandwidth and guarantees the control performance. Concurrent learning is used to estimate unknown rotor failures and provides failure information to NMPC, which enables the system to handle failure autonomously without any information in priori. Simulation results show that, compared with cascaded MPC, the proposed method can autonomously identify the failure and track the desired trajectory with an unknown complete rotor failure.

There is still some work to be done in the future. Firstly, the simulation is implemented on the PC, which ignores the influence of computational complexity. Secondly, the influence of disturbance and modeling uncertainty is not considered. Finally, the tracking error is large before the FDD is

fully converged. In future work, we will focus on the deployment of the method on embedded platforms and improve the robustness to further enhance the practicality of the method. And we will also investigate the application of this method in more complex failure situations, such as the simultaneous failure of multiple actuators.

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